

Lab_Day01-Cybersecurity Fundamentals

Lab Objectives

By the end of this lab session, students will be able to:

- Configure Hypervisor Environments: Deploy and manage a Type-2 hypervisor (Oracle VirtualBox) including core platform packages and extensions necessary for kernel-level guest integration.
- Administer Advanced Package Management: Utilize the Debian-based package management system (apt) to dynamically update source repositories, execute full-system upgrades, and handle dependencies via the command-line interface (CLI).
- Deconstruct Vulnerability Severity Frameworks: Analyze security advisories within the National Vulnerability Database (NVD) and dissect individual component vectors of the Common Vulnerability Scoring System (CVSS v4.0).
- Understand Threat Models and Quantitative Risks: Contrast theoretical system vulnerabilities against practical operational environments using the FIRST CVSS Calculator to derive and justify Base vs. Environmental threat scores.

Lab Tasks

TASK 1: INSTALL VIRTUAL BOX	2
TASK 2: DOWNLOAD AND IMPORT KALI LINUX VIRTUAL MACHINE IMAGE IN VIRTUALBOX.....	2
TASK 3: START KALI LINUX AND INITIAL CONFIGURATION	2
TASK 4: THREAT MODELING & CVSS CALCULUS.....	2

Task 1: Install Virtual Box

Install VirtualBox virtualization software by downloading and Installing from official website: <https://www.virtualbox.org/wiki/Downloads>

- Use appropriate VirtualBox Platform Package for your machine
- Also install VirtualBox Extension Pack

Task 2: Download and Import Kali Linux Virtual Machine Image in VirtualBox

Download the pre-built Kali Linux virtual machine image optimized for VirtualBox.

- Navigate to Kali Linux download page and download VirtualBox premade image: <https://www.kali.org/get-kali/#kali-virtual-machines>
- Unzip and Import image in VirtualBox following instructions: <https://www.kali.org/docs/virtualization/import-premade-virtualbox/>

Task 3: Start Kali Linux and Initial Configuration

Boot Kali Linux virtual machine and perform initial system updates.

- Start the virtual machine, Select "Kali Linux" in the VirtualBox Manager and click "Start".
- Wait for Kali Linux to boot (may take 1-2 minutes on first boot)
- Log in to Kali Linux:
 1. Default Username: `kali`
 2. Default Password: `kali`
 3. Open Terminal by Clicking on the terminal icon in the top-left corner (black box icon), Or press `Ctrl + Alt + T`
 4. Update Kali Linux system: `sudo apt update`
- Upgrade all packages: `sudo apt full-upgrade -y` (This may take 10-30 minutes depending on your internet speed; Press "Y" or Enter if prompted)

Next 2 steps will be needed when we will attempt offensive security labs.

- Install OpenVPN (required for TryHackMe connection): `sudo apt install openvpn -y`
- Verify OpenVPN installation: `openvpn --version` (You should see version information displayed)

Task 4: Threat Modeling & CVSS Calculus

Objective: Understanding how a Vulnerability (bug) turns into a Risk (probability * impact) based on the Threat environment.

Scenario: Your university hosts a self-hosted Git repository containing grading scripts. A remote code execution flaw is announced, but the repository is **strictly behind a campus VPN** requiring multi-factor

authentication (MFA). Navigate to the NVD (National Vulnerability Database) to look up a vulnerability (CVE-2026-3854).

- Check CVSS v4.0 score
- Note Attack Vector that gets this score.
- Calculate same score using CVSS Calculator: <https://www.first.org/cvss/calculator/4.0>
- Further using the FIRST CVSS (Common Vulnerability Scoring System) Calculator, compute two scores:
 - **Base Score:** The inherent severity of the vulnerability.
 - **Environmental Score:** Modify the metrics to fit the university scenario (e.g., changing Attack Vector from Network to Adjacent, adjusting Modified Integrity Impact due to the VPN/MFA layer).

You will understand that *a "Critical" vulnerability doesn't always equal a "Critical" risk if proper environmental controls are in place.*
